AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

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Roll No. :	64
Experiment No.:	**6**
Title of the Experiment:	**Develop and Evaluate Regression Models using Statistical Performance Measures**
Date of Performance:	11/08/26
Date of Submission:	18/08/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :

Aim: To develop regression models for predicting a continuous variable using the Pima Indians Diabetes Dataset and evaluate their performance using appropriate statistical metrics.

Objectives

After completing this experiment, the student will be able to:

1. Develop regression models to predict a continuous health-related variable from medical attributes.
2. Evaluate and interpret the performance of regression models using appropriate statistical measures.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains medical information for **768 female patients**.

For this experiment, **BMI may be selected as the continuous target variable**, while other relevant medical attributes are used as input features.

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select a continuous target variable and relevant predictor variables.
3. Check and preprocess missing or invalid values.
4. Divide the dataset into training and testing sets.
5. Develop a suitable regression model.
6. Train the model using the training data.
7. Generate predictions for the test data.
8. Evaluate the model using appropriate performance metrics.
9. Analyze residuals and compare actual and predicted values.

10. Interpret the model results and identify possible limitations.

Description of the Experiment

Regression is a supervised learning technique used to predict a continuous numerical output.

The experiment follows: Dataset, Select Features and Target, Preprocess Data, Split Data, Train Regression Model, Generate Predictions, Evaluate Performance and Interpret Results

Students develop a regression model to estimate a continuous health-related variable based on other available attributes.

Detailed Description of Statistical Methods

Linear Regression

A simple linear regression model can be represented as:

$$y = \beta_0 + \beta_1 x + \epsilon$$

For multiple predictors:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

where:

- y= target variable
- x_i= predictor variables
- β_i= model coefficients
- ϵ= error term

Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

It represents the average absolute prediction error.

Mean Squared Error

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Larger errors are penalized more heavily.

Root Mean Squared Error

$$RMSE = \sqrt{MSE}$$

RMSE is expressed in the same units as the target variable.

Coefficient of Determination

$$R^2 = 1 - \frac{\Sigma(y_i - \hat{y}_i)^2}{\Sigma(y_i - \bar{y})^2} R^2$$

indicates the proportion of variation in the target variable explained by the model.

Residual Analysis

A residual is:

$$e_i = y_i - \hat{y}_i$$

Residual analysis helps identify:

- Systematic prediction errors
- Nonlinear patterns
- Unequal variance
- Potential outliers

Implementation:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score


# Load dataset

url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv "columns = [

    "Pregnancies", "Glucose", "BloodPressure",

    "SkinThickness", "Insulin", "BMI",

    "DiabetesPedigreeFunction", "Age", "Outcome"

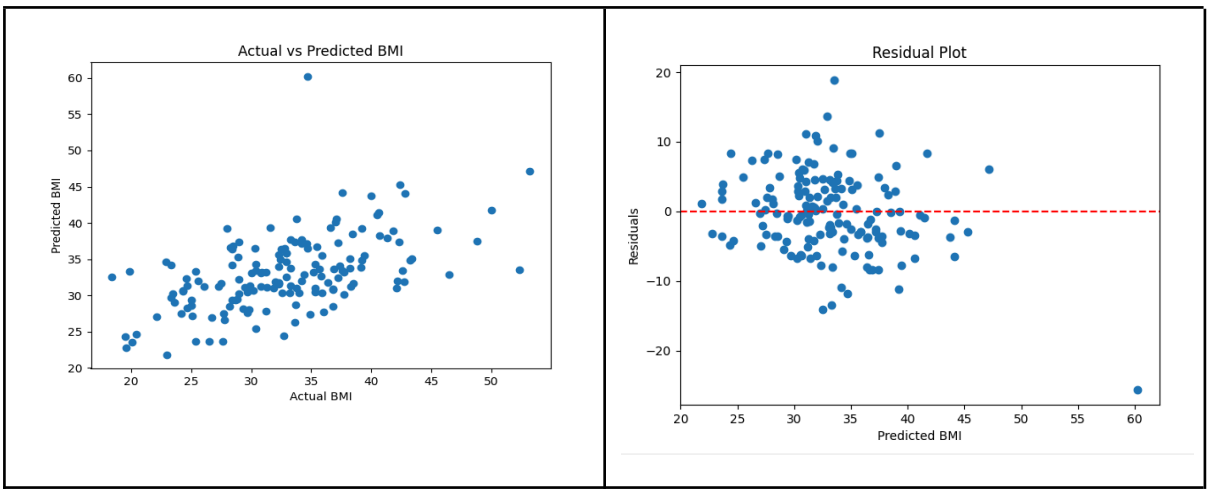
]

data = pd.read_csv(url, names=columns)
```

```
print("First 5 rows:")
print(data.head())
# Replace invalid 0 values with median
cols = ["Glucose", "BloodPressure", "SkinThickness", "Insulin", "BMI"]
for col in cols:
    data[col] = data[col].replace(0, np.nan)
    data[col] = data[col].fillna(data[col].median())
# Select input and target
X = data.drop("BMI", axis=1)
y = data["BMI"]
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Create and train model
model = LinearRegression()
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Performance measures
MAE = mean_absolute_error(y_test, y_pred)
MSE = mean_squared_error(y_test, y_pred)
RMSE = np.sqrt(MSE)
R2 = r2_score(y_test, y_pred)
print("\nPerformance Metrics:")
print("MAE  :", MAE)
print("MSE  :", MSE)
print("RMSE :", RMSE)
print("R2   :", R2)
# Actual vs Predicted
```

```
print("\nActual vs Predicted BMI:")
for actual, predicted in zip(y_test.head(10), y_pred[:10]):
    print("Actual:", round(actual, 2),
          "Predicted:", round(predicted, 2))
# Residual plot
residuals = y_test - y_pred
plt.scatter(y_pred, residuals)
plt.axhline(0, color="red", linestyle="--")
plt.xlabel("Predicted BMI")
plt.ylabel("Residuals")
plt.title("Residual Plot")
plt.show()
# Actual vs Predicted plot
plt.scatter(y_test, y_pred)
plt.xlabel("Actual BMI")
plt.ylabel("Predicted BMI")
plt.title("Actual vs Predicted BMI")
plt.show()
```

Output:



```
PS D:\65 exp> python exp6.py
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1

Missing values:
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Performance Metrics:
MAE  = 4.47529731767549
MSE  = 33.917131715253326
RMSE = 5.823841662962114
R2   = 0.2245296074764075

Actual vs Predicted:
   Actual BMI  Predicted BMI
0        34.0      30.363728
1        35.7      33.689430
2        30.8      31.250070
3        24.6      32.348081
4        29.9      31.307778
5        37.7      30.183501
6        20.4      24.621205
7        33.8      30.963086
8        31.3      31.130530
9        33.7      28.672292
PS D:\65 exp>
```

Conclusion

A regression model was developed to predict a continuous health-related variable using the Pima Indians Diabetes Dataset. The model was evaluated using appropriate regression metrics, and residual analysis was performed to understand the model's prediction errors and limitations.

Questions to be Answered by Students

1. Which variable was selected as the target, and why was it suitable for regression?

Ans: Target Variable: BMI was selected because it is a continuous numerical variable suitable for regression.

2. What are the values of MAE, MSE, RMSE, and R^2obtained by the model?

Ans:MAE=4.475297

MSE = 33.9171

RMSE = 5.8238

R^2 = 0.2245

3. Which performance metric provides the most useful interpretation for this problem? Justify your answer.

Ans: Best Metric: RMSE, because it is in the same unit as BMI and shows the prediction error.

4. What patterns or issues were observed from the residual analysis?

Ans: Residual Analysis: Residuals were mostly scattered around zero, with some larger errors/outliers.

5. State **two limitations** of the developed regression model.

Ans:

- Linear Regression assumes a linear relationship between the input variables and BMI, so it may not capture complex nonlinear relationships.
- The model is sensitive to outliers, and extreme medical observations can affect the regression coefficients and predictions.